Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Quantum pure states are not enough

Files.

- File: `briefs/PureStatesNotEnough/PureStatesNotEnough.lean`
- Library: `PureStatesNotEnough`
- Imports: two new `ForMathlib` files:
  `ForMathlib/Analysis/Convex/Combination.lean`: a map that preserves the convex combinations of two points of a convex set preserves all finite convex combinations (`Convex.map_sum_of_map_add`), and a convex set contains them (`Convex.sum_mem_of_semifield`), with scalars in any ordered semifield such as $\mathbb{R}_{\geq 0}$.
  `ForMathlib/MeasureTheory/Constructions/HaarToSphere.lean`: Archimedes' hat-box theorem. The map $(z, \theta) \mapsto (\sqrt{1-z^2}\cos\theta, \sqrt{1-z^2}\sin\theta, z)$ is injective on $(-1, 1) \times (-\pi, \pi]$ with image the sphere minus its poles and area Jacobian 1 (`euclideanHausdorffMeasure_image_cylinderToSphere`, by TauCeti's area formula), so Mathlib's surface measure is $dz\, d\theta$ (`Measure.map_toSphere_volume_eq_map_cylinderToSphere`) and the height is distributed as $2\pi\, dz$ (`Measure.map_toSphere_volume_height`).
  Four new `ForPhyslib` files on the qubit, in `ForPhyslib/QuantumMechanics/Qubit/`:

`BlochSphere.lean`: the Bloch point of a ray (`blochPoint`), injective, onto the unit sphere, and a homeomorphism `blochSphere` : $P(\mathbb{C}^2) \simeq S^2$; $|\psi\rangle\langle\psi| = \frac{1}{2}(1 + a \cdot \sigma)$ (`hermitianProjector_eq`).
`BlochRotation.lean`: a unitary U acts on Bloch points by TauCeti's rotation `pauliRotation` U (`blochPoint_unitaryAction`); every rotation is the action of an element of $SU(2)$ (`exists_unitaryAction_eq`); two unitaries act alike iff they differ by a phase (`unitaryAction_eq_iff`); complex conjugation of the components is the reflection $a_1 \mapsto -a_1$ (`blochPoint_conjAction`) and no unitary action (`conjAction_ne_unitaryAction`).
`AntisymplecticReflection.lean`: complex conjugation `conjSphere` of the Riemann sphere is smooth, preserves the Fubini–Study density (`fubiniStudyDensity_conj`), pulls the Fubini–Study form back to its negative (`mpullback_conjSphere`), is no time-t map of a Hamiltonian flow (`globalFlow_ne_conjSphere`), and corresponds to `conjAction` (`equivProjectivization_conjSphere`).
`Precession.lean`: every Hermitian H is $c + \frac{1}{2} n \cdot \sigma$ (`exists_eq_smul_one_add_pauli`); along a solution of $i\psi' = H\psi$ the Bloch vector satisfies $a' = n \times a$ (`hasDerivAt_blochVector`).
These use TauCeti's Pauli map, double cover $SU(2) \to SO(3)$, Hermitian projectors, topology of projective space, Fubini–Study form on `OnePoint` $\mathbb{C}$, global Hamiltonian flows, area formula and identification of `Measure.toSphere` with Hausdorff measure.
Existing files: `ForMathlib/Analysis/InnerProductSpace/l2Space.lean` (`Orthonormal.countable`), `ForPhyslib/QuantumMechanics/DensityOperator.lean` (`IsDensityOperator`, `posTrace`), `ForPhyslib/QuantumMechanics/Wigner/Density.lean` (`IsDensity`, `proj`) and `ForPhyslib/ProbabilisticTheory/HilbertSpace/TraceClass/CanonicalExpansion.lean` (`IsTraceClass.exists_hasSum_rankOne`).

Setting. The ensembles of CST over a measurable space X with Liouville measure μ are the probability measures $\nu \ll \mu$ (`classicalEnsembles` μ). Their mixtures are $a\nu_1 + b\nu_2$ with $a, b \in \mathbb{R}_{\geq 0}$, $a + b = 1$. `HasManyDisjointSets` μ says that for every N there are N disjoint measurable sets of positive finite measure. The density operators of QE-DENS are `IsDensity` (positive, trace one) on a finite-dimensional space and `IsDensityOperator` (positive, trace class, trace one) on a separable complex Hilbert space $\mathcal{H}$ of any dimension. The pure states (`pureStates`) are the projections $|\psi\rangle\langle\psi|$ with $\|\psi\| = 1$, a copy of $P(\mathcal{H})$ inside the bounded operators, with the Borel σ-algebra of the operator norm. An identification of two ensemble spaces is modelled as a map that preserves mixtures of two ensembles.

Results.

Brief	Lean
CST, the ensemble space	`classicalEnsembles`; it is convex: `Convex.classicalEnsembles`
(used for Theorem 3) a Darboux chart has N disjoint sets of positive finite Liouville measure for every N	`hasManyDisjointSets_restrict`: any nonempty open set in a nontrivial finite-dimensional real normed space, for any measure that is positive on open sets and finite on compact sets
(used for Theorem 3) the CST ensemble space is infinite-dimensional	`not_injOn_of_map_add`: no mixture-preserving map from it into a finite-dimensional real vector space is injective
Theorem 3, CST and QE-DENS are inconsistent	`cst_qeDens_inconsistent`, for $\dim \mathcal{H} < \infty$
Theorem 1, QP-RAY and CST are consistent iff $1 \leq \dim V < \infty$	not formalized; only its edges, below
(Theorem 1) $\dim V = 0$	`isEmpty_projectivization`: $P(V)$ is empty; `nonempty_projectivization_iff`
(Theorem 1) $\dim V = 1$	`subsingleton_projectivization`: $P(V)$ is a single point
(Theorem 1) $2 \leq \dim V$	`nontrivial_projectivization_iff`: in finite dimension $P(V)$ has two points iff $2 \leq \dim V$
(Theorem 3 at $\dim \mathcal{H} = 1$) a point with Liouville measure δ	`not_hasManyDisjointSets_dirac`; `cst_qeDens_consistent_of_finrank_one`: an injective mixture-preserving map exists
(Theorem 1, "only if") $P(\ell^2)$ versus the sphere	`mk_projectivization`: $P(V)$ has $\mathfrak{c}$ points; `nonempty_projectivization_equiv`: $P(\ell^2) \simeq P(\mathbb{C}^2)$ as sets
Remark after Example 2, the Born rule is not $\delta_{\psi\phi}$	`born_ne_delta`: e_0 and $(e_0 + e_1)/\sqrt{2}$ are distinct rays with Born probability $\frac{1}{2}$
Theorem 5, the map $r : M^1(P(\mathcal{H})) \to D(\mathcal{H})$	`barycenter` $r(\mu) = \int \rho \, d\mu(\rho)$; `isDensityOperator_barycenter`, `posTrace_barycenter`; `toDensity`
Theorem 5, r is affine	`barycenter_add`
Theorem 5, $D(\mathcal{H}) \equiv M^1(P(\mathcal{H}))/\sim$	`toDensity_surjective`, `densityEquiv`
Remark after Theorem 5, a class records the different preparations of one ensemble	`exists_barycenter_eq_and_ne`: $\frac{1}{2}\delta_{e_0} + \frac{1}{2}\delta_{e_1}$ and $\frac{1}{2}\delta_{e_+} + \frac{1}{2}\delta_{e_-}$ have the same barycenter but give e_0 different probabilities

Brief	Lean
Example 2, $P(\mathbb{C}^2)$ is a two-sphere	`nonempty_homeomorph_sphere`, from `blochSphere`: a homeomorphism onto the unit sphere of $\mathbb{R}^3$
Example 2, Darboux coordinates z, θ_{xy}, the Liouville measure is the area	`toSphere_eq_dz_dtheta`: the surface measure is $dz\,d\theta$ on $(-1,1)\times(-\pi,\pi]$; the height is distributed as $2\pi\,dz$
Example 2, Hamiltonian evolutions are the volume-preserving diffeomorphisms	`conjSphere_preserves_area_reverses_form`, `globalFlow_ne_conjSphere`: complex conjugation of the Riemann sphere preserves the area but is not Hamiltonian (see "Problems found")
Example 2, quantum evolutions are exactly the rotations	`exists_unitaryAction_eq_iff`: a map of rays is the action of a unitary iff it is a rotation on Bloch vectors; `unitaryAction_eq_iff`: unitaries act alike iff they differ by a phase
Example 2, linear Hamiltonians $h = n \cdot x$ give rotations about n	`exists_blochVector_precession`: quantum side only; every Hamiltonian $c + \frac{1}{2} n \cdot \sigma$ gives $a' = n \times a$

Differences from the brief.

- Theorem 3 compares the whole ensemble spaces, not those supported on a two-dimensional subspace A (see "Problems found"). It assumes $\dim \mathcal{H} < \infty$, which the brief derives from Theorem 1. Only the finite dimension of the real vector space of operators is used, not positivity or trace. "The spaces correspond" becomes a map from the CST ensembles to the density operators that is injective and preserves mixtures of two ensembles. The proof normalizes μ on $\dim + 2$ disjoint sets, takes an affine dependence among the images, and separates its positive and negative parts into two different mixtures with the same image.

- The Liouville measure enters only through `HasManyDisjointSets`, which holds on every Darboux chart. The symplectic structure itself is not used.

- Theorem 5 is proved for every separable complex Hilbert space, not only in finite dimension. $r(\mu)$ is the Bochner integral of the pure states in the operator norm. It is positive because the positive operators form a closed convex cone, and it has trace one by Parseval and monotone convergence over a countable Hilbert basis. r is onto by the canonical expansion $\rho = \sum_i \lambda_i |e_i\rangle\langle e_i|$, represented by $\sum_i \lambda_i \delta_{e_i}$. No Choquet theory is used. "$\equiv$" is the bijection between the quotient by $r(\mu_1) = r(\mu_2)$ and $D(\mathcal{H})$.

- Example 2 is checked at the level of topology and measure, and its symplectic part on the Riemann sphere `OnePoint` $\mathbb{C}$ with TauCeti's Fubini–Study form. The sphere is the unit sphere of $\mathbb{R}^3$ with Mathlib's surface measure, which TauCeti identifies with the Hausdorff area; "Darboux coordinates" becomes the statement that this measure is $dz\,d\theta$. The area-preserving map that is not Hamiltonian is complex conjugation of the Riemann sphere, which corresponds to the Bloch reflection $a_1 \mapsto -a_1$; area preservation is invariance of the Fubini–Study density, since there is no measure on `OnePoint` $\mathbb{C}$ yet. Quantum evolution is the action of a unitary on rays, and for a Hamiltonian the Schrödinger equation is a hypothesis on ψ; the precession $a' = n \times a$ is not integrated to a finite rotation.

- Not formalized:

  – Theorem 1;

– in Example 2: the Liouville measure of the Fubini–Study form and the symplectic form $dz \wedge d\theta$ in the chart; the diffeomorphism $\mathbb{C}P^1 \cong S^2$ with the comparison of the Fubini–Study and round area forms; ensembles as densities $\rho(z, \theta)$ (Radon–Nikodym, not restated); the classical side of linear Hamiltonians (moment maps);
– Insight 4, which is informal;
– that the pure states are the extreme points of $D(\mathcal{H})$, which the proof of Theorem 5 does not need.

Problems found in the brief. Theorem 5 holds, and Theorem 3 holds for $\dim \mathcal{H} \geq 2$, but their proofs have gaps. Example 2 holds once "volume-conserving" is read as "area-form-preserving". At $\dim \mathcal{H} = 1$ Theorem 3 depends on a convention of Theorem 1. Theorem 1 is not checked.

- Theorem 1 states $1 \leq \dim V$, while its proof uses $2 \leq \dim X$. For $\dim V = 1$ the state space $P(V)$ is a single point (`subsingleton_projectivization`), a manifold of dimension 0. So either 0-dimensional symplectic manifolds count, and the proof's bound should be $0 \leq \dim X$, or they do not, and the theorem should say $2 \leq \dim V < \infty$. The choice also decides Theorem 3. If a point counts, then for $\dim \mathcal{H} = 1$, which QE-DENS allows, the CST ensembles of the point are $\{\delta\}$ and the density operators are $\{1\}$, and the constant map identifies the two (`cst_qeDens_consistent_of_finrank_one`): Theorem 3 fails. The Lean proof of Theorem 3 avoids this through `HasManyDisjointSets`, which fails for a point (`not_hasManyDisjointSets_dirac`). So either Theorem 1 says $2 \leq \dim V$, or Theorem 3 needs $\dim \mathcal{H} \geq 2$.

- For $\dim V = 0$, $P(V)$ is empty (`isEmpty_projectivization`), and the empty set is a symplectic manifold of every dimension. Either lower bound, 1 or 2, needs CST to exclude it, for instance by requiring an ensemble: there is no probability measure on the empty set.

- The "only if" of Theorem 1 needs the identification of X with $P(V)$ to respect the topology. As sets, $P(\ell^2)$ and $P(\mathbb{C}^2)$, the sphere, are in bijection: both have $\mathfrak{c}$ points (`nonempty_projectivization_equiv`). Carrying the symplectic structure of the sphere over along such a bijection would make $P(\ell^2)$ a symplectic manifold. The theorem holds when V is Hausdorff and X has the quotient topology of $P(V)$; for a non-Hausdorff V, $P(V)$ need not be Hausdorff even in finite dimension, and the "if" fails. "Since X is a symplectic manifold, $\dim X < \infty$" holds only for finite-dimensional symplectic manifolds, and the projective space of an infinite-dimensional Hilbert space is an infinite-dimensional one. The finiteness comes from CST itself: the Liouville measure ω^n needs a finite n.

- In the proof of Theorem 3, for $\dim \mathcal{H} \geq 3$ the sphere $P(A)$ has dimension $2 < 2(\dim \mathcal{H} - 1)$ and therefore Liouville measure zero. No CST ensemble has support in $P(A)$, so the classical side is empty, not infinite-dimensional. The argument works as written only for $\dim \mathcal{H} = 2$. Comparing the whole spaces repairs it.

- The proof of Theorem 5 cites Choquet theory, which concerns compact convex sets, but $D(\mathcal{H})$ is not compact when $\dim \mathcal{H} = \infty$. The proof also never shows that every density operator is represented, which $D(\mathcal{H}) \equiv M^1/\sim$ needs. Both gaps are closed by the canonical expansion. "Every continuous linear functional $f : D(\mathcal{H}) \to \mathbb{R}$" should read continuous affine function, since $D(\mathcal{H})$ is not a vector space.

- Example 2: area-preserving diffeomorphisms of the sphere include the orientation-reversing ones, such as $z \mapsto -z$, which are not Hamiltonian. So Hamiltonian evolution preserves the area

form, not only the area. Checked in Lean for complex conjugation of the Riemann sphere, the reflection $a_1 \mapsto -a_1$ of the Bloch sphere: it preserves the Fubini–Study density but reverses the form, and no time-t map of a Hamiltonian flow equals it (`conjSphere_preserves_area_reverses_form`, `globalFlow_ne_conjSphere`).

- CST says $p(\xi_1 \mid \xi_2) = 0$ "for all $\xi_1, \xi_2 \in X$", which includes $p(\xi \mid \xi) = 0$. It should say $\xi_1 \neq \xi_2$, as the Remark after Example 2 ($p(\psi \mid \phi) = \delta_{\psi\phi}$) has it.
- Typos: "non necessarily" should be "not necessarily"; "r defined about" should be "r defined above".